

Hyperfine selective collisional excitation of interstellar molecules

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Summary. We discuss in detail the hyperfine selective collisional excitation of linear rotor and symmetric top molecules and show that the commonly used simpler approximations for handling the hyperfine selective collisional excitation are not sufficient. The scattering process is described in the framework of formal scattering theory, which also allows for a concise derivation of the CS and IOS approximations. We give numerical results for the scattering of para NH_3/He and HCN/He . The correct description of the hyperfine selective collisional excitation is important for the interpretation of radioastronomical observations of these molecules, which show that the hyperfine states are not thermally populated in many molecular clouds.

Key words: molecules – collision processes

1. Introduction

Of the most common constituents of interstellar molecules, H , ^{12}C , ^{14}N , and ^{16}O , the ^{14}N nucleus is the only one with a nonvanishing quadrupole moment. This causes nuclear quadrupole hyperfine splitting in the spectrum of several of the molecules of astrophysical interest. The quadrupole coupling constant usually is of the order of several MHz for an end ^{14}N -nucleus, so that the separation of the hyperfine components of transitions in the cm- and mm-wavelength range is several km s^{-1} on an equivalent Doppler-velocity scale. The hyperfine structure is, therefore, well resolved in most sources, which are not too hot and which have fairly simple velocity structure.

Resolvable hyperfine structure is important to the astrophysical interpretation of molecular observations because the optical depth of the observed transition can be obtained by comparing the relative intensity of the hyperfine transitions. For optically thin lines the intensity ratio is proportional to the ratio of the corresponding transition strengths and approaches unity for optically thick lines. However, this is only valid if all hyperfine levels have the same excitation temperature.

Observations of several molecules indicate intensities of the hfs-transitions being in conflict with the above assumption. For example, HCN shows anomalous hfs-intensities in the $J=1 \rightarrow 0$ rotational transition in several warm (see Turner et al., 1977, and references therein) and cold clouds (Walmsley et al., 1982). Corresponding anomalies have been found for HC_3N (Mac Gee et al., 1973; Churchwell et al., 1977; Tölle et al., 1981).

For NH_3 , anomalies in the $(J, K)=(1, 1)$ inversion transition were first detected by Matsakis et al., 1977, in absorption towards the continuum source DR21, but this seemed to be a rather special phenomenon connected with the special geometry of that source. With the increasing sensitivity of new receiver systems, recent observations show similar anomalies in the emission spectra from several other clouds (Stutzki et al., 1982; Batrla et al., 1983; Morane, 1984). A systematic search for $\text{NH}_3(1, 1)$ anomalies with high signal/noise ratio reveals this to be a rather common phenomenon, occurring in around 50% of the medium warm molecular clouds, i.e., those with rotational temperatures in the range 15–30 K (Stutzki et al., 1984).

For the interpretation of observed spectra with intensity anomalies in the hyperfine structure it is necessary to understand the excitation mechanisms in the molecular clouds with the help of appropriate radiative transfer models. Selective trapping effects in the hyperfine transitions, as well as Doppler and thermal overlap effects have been shown by several authors to be important in explaining the observed anomalies (Kwan and Scoville, 1975; Gottlieb et al., 1975; Matsakis et al., 1977; Guilleoteau and Baudry, 1981). These radiative processes are balanced by the collisions, so that the resulting strength of the anomalies is strongly influenced by the hyperfine selective collision rates. All previous investigations use rather simple models for the collisional excitation of the hyperfine states.

In this paper we discuss in detail the collisional excitation of molecular hyperfine states starting from a general description by means of the transition matrix. As a result, we can show that the models commonly used are too simple. We give numerical results for the hyperfine selective collision cross sections for NH_3 and HCN . The strength of the anomalies is determined by the balancing between radiative processes and the collisional rates, which in turn are proportional to the H_2 density. Radiative transfer calculations show that the interpretation of the anomalous spectra gives an independent way for determining the H_2 -densities in molecular clouds. The resulting interpretation of the observed anomalies for NH_3 is presented in a separate paper (Stutzki and Winnewisser, 1984).

For hyperfine levels, which have the same excitation temperature, the population is proportional to the statistical weight of the hyperfine levels (neglecting the hyperfine energy splitting, which is orders of magnitude lower than typical excitation energies). In this case we express the population in level x , J , F

$$n_{xJF} = \frac{(2F+1)}{(2I+1)(2J+1)} n_{xJ}, \quad (1)$$

where J is the rotational angular momentum, I the nuclear spin and F the total angular momentum of the molecule; x labels

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further internal quantum numbers, e.g., the projection K of J on the molecular symmetry axis in case of a symmetric top or the inversion states of ammonia. Let us denote by $\sigma_{a';a}$ the collisional cross section for a transition from state a to a' , with $C_{a';a} = \langle \sigma_{a';a} v \rangle$ the collisional rate constants and with $R_{a';a} = n C_{a';a}$ the corresponding collision rates. In the rate equations determining the population of the states, the term describing collisional excitation of the hyperfine state $x'J'F'$ then contains the sum of transitions out of the hyperfine states F of the initial rotational levels J , x

$$\sum_F n_{xJF} R_{x'J'F';xJF} = \frac{n_{x,J}}{(2I+1)(2J+1)} \sum_F (2F+1) R_{x'J'F';xJF}. \quad (2)$$

Equation (1) has been used in deriving (2), and therefore it is valid only in the case of equal excitation temperature. As shown below [Eq. (12b)] the F -sum can be expressed simply by the collision rate for the rotational transition $R_{x'J';xJ}$, so that (2) results in

$$\sum_F n_{xJF} R_{x'J'F';xJF} = \frac{n_{x,J}}{(2I+1)(2J+1)} R_{x'J';xJ}. \quad (3)$$

A similar equation may be derived for the deexcitation of a specific hyperfine state. Thus, in the case of equal excitation temperature in all hyperfine levels, the excitation processes are only governed by the rotational collision rates. When the hyperfine states are not populated thermally, however, detailed knowledge of the hyperfine selective collisional processes is necessary for determining the population of the hyperfine states.

2. Calculation of hyperfine selective collision rates

Let us consider the transition operator $T(E)$ determined by the Lippmann-Schwinger equation (Goldberger and Watson, 1964; Joachain, 1975)

$$T(E) = V + V \frac{1}{E - H_0 + i\eta} T(E), \quad (4)$$

where V is the operator of the interaction potential and H_0 is the Hamilton operator of the collision partners neglecting the interaction: $H_0 = H - V$. It is most convenient to describe the scattering process in the center of mass system. Let us assume, that the collision partner of the molecule of interest is a structureless particle (He-atom, H_2 -molecule in the rotational ground state). The quantum numbers specifying the initial and final states of the collision system then are: 1) the magnitude of the momentum of the molecule $\hbar k$, giving the collision energy $(\hbar^2/2\mu)k^2$, where μ is the reduced mass, 2) the orbital angular momentum of the relative motion l , 3) the rotational angular momentum of the molecule J , 4) the nuclear spin of the ^{14}N -nucleus I , 5) the total angular momentum of the molecule $F = J + I$, 6) the total angular momentum of the collision system $L = F + l$ and any additional quantum numbers x describing the internal excitation of the molecule (e.g., for a symmetric top the projection K of J on the molecular symmetry axis or the inversion quantum numbers in the case of ammonia).

In the basis $|a\rangle = |k, x, ((J, I)F, l)LM_L\rangle$, the transition matrix elements are, of course, diagonal in the magnitude of the nuclear spin I and, because of the rotational invariance of the total Hamilton operator ($[H, L] = 0$), are diagonal in L , M_L and independent of M_L :

$$\begin{aligned} \langle k'x'((J', I)F', l')L'M_L' | T(E) | kx((J, I)F, l)LM_L \rangle \\ = T_{k'x'J'F'l';kxJF l}^{L, I(E)} \delta_{L'L} \delta_{M_L' M_L}. \end{aligned}$$

The collision cross sections then are given by (see Goldberger and Watson, 1964)

$$\begin{aligned} \sigma_{x'J'F';xJF}(k) &= \left(\frac{2\pi}{\hbar} \right)^4 \mu^2 \frac{1}{k} \frac{1}{2F+1} \\ &\sum_L (2L+1) \sum_{l'l'} \frac{1}{4\pi} |T_{k'x'J'F'l';kxJF l}^{L, I}|^2, \end{aligned} \quad (5)$$

where $T_{a';a} = \langle a' | T(E = E_{a'} = E_a) | a \rangle$ is the transition matrix element reduced "on the energy shell", and $E_{k,x,J,F} = (\hbar^2/2\mu)k^2 + \varepsilon_{x,J,F}$ with the excitation energy $\varepsilon_{x,J,F}$ of the molecule.

The interaction potential is determined by the electronic structure of the colliding molecules and is not influenced by the orientation of the nuclear spin deep inside the molecule. This is especially true for the low collision energies relevant under astrophysical conditions in molecular clouds. To take advantage of this in the calculations, it is convenient to change to a representation, where the nuclear spin is decoupled from the other quantum numbers. This is achieved by recoupling the three angular momenta J , I and l from the coupling scheme $|((J, I)F, l)LM_L\rangle$ to a coupling scheme $|((J, l)\tilde{L}, I)LM_L\rangle$ and then decoupling I from $\tilde{L}$, thus arriving at

$$\begin{aligned} |(J, l)\tilde{L}M_{\tilde{L}}\rangle |IM_I\rangle &= \sum_{LM_L} (-1)^{I+F-M_L} [LF\tilde{L}] \\ &\begin{pmatrix} \tilde{L} & I & L \\ M_{\tilde{L}} & M_I & M_L \end{pmatrix} \begin{Bmatrix} I & J & F \\ l & L & \tilde{L} \end{Bmatrix} |((J, I)F, l)LM_L\rangle \end{aligned} \quad (6)$$

(here $[ab]$ is an abbreviation for $(2a+1)(2b+1)$, $\begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix}$ and $\begin{Bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{Bmatrix}$ are the Wigner 3j- and 6j-symbols in the notation and phase convention as defined by Edmonds, 1960).

The potential matrix then is diagonal and independent of I and M_I , i.e.,

$$\begin{aligned} \langle k'x'(J'l)\tilde{L}'M_{\tilde{L}}' | \langle IM_I' | V | IM_I \rangle | kx(Jl)LM_{\tilde{L}} \rangle \\ = \delta_{M_I' M_I} \delta_{\tilde{L}' \tilde{L}} \delta_{M_L' M_L} \langle k'x'J'l' | V^{\tilde{L}} | kxJl \rangle, \end{aligned}$$

where the reduced matrix elements are calculated only between rotational states and again use has been made of the diagonality in $\tilde{L}$, $M_{\tilde{L}}$ due to the rotational symmetry. The angular dependence of the interaction potential may be expanded in spherical harmonics in a molecule fixed coordinate system with the z -axis lying in the symmetry axis of the molecule (here assumed to be a symmetric top)

$$V(R, \theta, \phi) = \sum_{\lambda\mu} v_{\lambda\mu}(R) Y_{\lambda\mu}(\theta, \phi).$$

The potential matrix elements between rotational states of the molecule then may be readily derived (compare Green, 1976). The result for a symmetric top molecule ($x = K$) is

$$\begin{aligned} \langle k'J'K'l' | V^{\tilde{L}} | kJKl \rangle &= \sum_{\lambda\mu} (-1)^{K+L+\lambda} \\ &\cdot v_{\lambda\mu}^{l'l'}(k', k) \left(\frac{[JJ'l'\lambda]}{4\pi} \right)^{1/2} \\ &\cdot \begin{pmatrix} l' & \lambda & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} J' & \lambda & J \\ K' & \mu & -K \end{pmatrix} \begin{Bmatrix} J' & \lambda & J \\ l & \tilde{L} & l' \end{Bmatrix}, \end{aligned} \quad (7)$$

where $v_{\lambda\mu}^{I' I}(k', k)$ is the matrix element over the radial wave function

$$v_{\lambda\mu}^{I' I}(k', k) = \frac{2}{\pi} (-i)^{I'} (i)^I \cdot \int R^2 dR j_{I'}(k'R) j_I(kR) v_{\lambda\mu}(R).$$

For a linear rotor (e.g., HCN) the potential expansion coefficients with $\mu \neq 0$ vanish and the corresponding matrix elements may be obtained by setting $K = K' = 0$ in (7) (Green, 1976).

Whereas the potential matrix is diagonal in M_I , M_I' and $\tilde{L}$, the Green operator in the Lippmann-Schwinger equation $1/(E - H_0 + i\eta)$, and therefore the transition matrix, are not diagonal, because the product states (6) are not eigenfunctions of H_0 . The resulting Lippmann-Schwinger equation [Eq. (4)] for the transition matrix elements is

$$\begin{aligned} T_{k'x'J'I'; kxJF}^{I(E)} \tilde{L}' M_I' M_I \tilde{L} M_I &= \langle k'x'J'I' | V^{\tilde{L}} | kxJF \rangle \delta_{M_I' M_I} \delta_{\tilde{L}' \tilde{L}} \delta_{M_I' M_I} \\ &\quad \frac{1}{(2\pi)^3} \int k''^2 dk'' \sum_{\substack{J'' M_I'' \\ \tilde{L}'' M_I''}} \langle k'x'J'I' | V^{\tilde{L}} | k''x''J''I'' \rangle \\ &\quad \sum_{FLM_L} [LF] ([\tilde{L}' \tilde{L}''])^{1/2} \begin{pmatrix} \tilde{L}' & I & L \\ M_I' & M_I & M_L \end{pmatrix} \begin{pmatrix} \tilde{L}'' & I & L \\ M_I'' & M_I' & M_L \end{pmatrix} \\ &\quad \begin{Bmatrix} I & J'' & F \\ I'' & L & \tilde{L}'' \end{Bmatrix} \begin{Bmatrix} I & J' & F \\ I' & L & \tilde{L}' \end{Bmatrix} \frac{1}{E - E_{k''x''J''I''} + i\eta} \\ &\quad \cdot T_{k''x''J''I''; kxJF}^{I(E)} \tilde{L}'' M_I'' M_I \tilde{L} M_I. \end{aligned}$$

For a molecule without a nuclear spin, i.e., $I=0$, $M_I=0$, the Wigner 6j-symbols reduce to Kronecker functions, so that $J''=F''$ and $\tilde{L}''=\tilde{L}'=L$; the sum over L, M , gives a Kronecker function for M_I' and M_I . The transition matrix then is diagonal in $\tilde{L}'$, $\tilde{L}$ and M_I' , M_I :

$$T_{\dots \tilde{L}' M_I' \tilde{L} M_I}^{I=0} = \delta_{\tilde{L}' \tilde{L}} \delta_{M_I' M_I} T_{\dots}^{\tilde{L}, I=0}$$

and the reduced transition matrix elements satisfy the Lippmann-Schwinger equation

$$\begin{aligned} T_{k'x'J'I'; kxJF}^{\tilde{L}, I=0(E)} &= \langle k'x'J'I' | V^{\tilde{L}} | kxJF \rangle \\ &\quad + \frac{1}{(2\pi)^3} \int k''^2 dk'' \sum_{x''J''I''} \langle k'x'J'I' | V^{\tilde{L}} | k''x''J''I'' \rangle \\ &\quad \cdot \frac{1}{E - E_{k''x''J''I''} + i\eta} T_{k''x''J''I''; kxJF}^{\tilde{L}, I=0(E)}. \end{aligned} \quad (8)$$

Neglecting the hyperfine energy splitting in the Hamilton operator H_0 , in the Lippmann-Schwinger equation for the hyperfine selective transition matrix elements, $E_{k''x''J''I''}$ is approximated by $E_{k''x''J''}$. Because of the orthogonality in the 6j-symbols the F -sum reduces to a Kronecker symbol for $\tilde{L}'$, $\tilde{L}$ and the L, M_L sum results in a Kronecker symbol for M_I' , M_I and M_I' , M_I . Thus the Green operator becomes diagonal in all these quantum numbers and therefore the transition matrix elements are also diagonal in $\tilde{L}$, M_I and M_L and independent of M_I' and M_I . The diagonal matrix elements are then solutions of the same Eqs. (8) as the matrix elements with $I=0$:

$$T_{\dots \tilde{L}' M_I' \tilde{L} M_I}^I \approx \delta_{M_I' M_I} \delta_{\tilde{L}' \tilde{L}} \delta_{M_I' M_I} T_{\dots}^{\tilde{L}, I=0}. \quad (9)$$

As the hyperfine energy splitting is always very small compared to the relevant transition energies (of rotational and even inversion transitions) and the collision energies (ranging from 10 K to some 100 K), neglecting the hyperfine energy splitting is an excellent approximation. The usual approximations made for collision

cross section calculations, which are discussed below, are much more radical. The main effect of neglecting the hyperfine energy splitting in the Green operator is that the detailed balance relations then are not accurately satisfied for collision energies very low compared to the hyperfine energy splitting.

Recoupling the angular momenta back to the total angular momentum representation by reversing (6) and approximating the transition matrix elements by (9) one gets

$$\begin{aligned} T_{k'x'J'F'; kxJF}^{L, I(E)} &= \sum_{\tilde{L}} (-1)^{F'+F+I'+I} [\tilde{L}] ([FF'])^{1/2} \\ &\quad \cdot \begin{Bmatrix} \tilde{L} & I & L \\ F' & I' & J' \end{Bmatrix} \begin{Bmatrix} \tilde{L} & I & L \\ F & I & J \end{Bmatrix} T_{k'x'J'I'; kxJF}^{\tilde{L}, I=0(E)} \end{aligned} \quad (10)$$

and for the collision cross section, following (5),

$$\begin{aligned} \sigma_{x'J'F'; xJF}(k) &= \left(\frac{2\pi}{h} \right)^4 \mu^2 \frac{k'}{k} \sum_{\tilde{L}} [\tilde{L} \tilde{L}'] \sum_L [L] \\ &\quad \cdot \begin{Bmatrix} \tilde{L} & I & L \\ F' & I' & J' \end{Bmatrix} \begin{Bmatrix} \tilde{L} & I & L \\ F & I & J \end{Bmatrix} \begin{Bmatrix} \tilde{L} & I & L \\ F' & I' & J' \end{Bmatrix} \begin{Bmatrix} \tilde{L} & I & L \\ F & I & J \end{Bmatrix} \\ &\quad \cdot \frac{1}{4\pi} T_{k'x'J'I'; kxJF}^{\tilde{L}, I=0*} T_{k'x'J'I'; kxJF}^{\tilde{L}, I=0}. \end{aligned} \quad (11)$$

The hyperfine selective transition matrix elements for one specific total angular momentum L , according to (10), are determined by the pure rotational transition matrix elements with total angular momenta $\tilde{L}$ ranging from $|L-I|$ to $L+I$. Therefore, the expression for the hyperfine selective collisional cross section (11) contains not only the absolute squares of rotational transition matrix elements, but also terms where the relative phases of the transition matrix elements of different total angular momenta interfere. Thus, there is no simple way of relating the hyperfine selective cross sections directly to the pure rotational cross sections.

The nuclear spin does not influence the collision dynamics. As a result, the hyperfine collision rates summed over the final states F' of a specific hyperfine multiplet simply give the corresponding rotational cross section

$$\sum_{F'} \sigma_{x'J'F'; xJF}(k) = \sigma_{x'J'; xJ}^{I=0}, \quad (12a)$$

as can be verified by applying the sum to the expression in (11). According to detailed balance, the total rate out of an hyperfine multiplet, obtained by averaging over the initial states F is simply the pure rotational cross section multiplied by the statistical weight of the final state

$$\begin{aligned} \sum_F \frac{(2F+1)}{(2J+1)(2I+1)} \sigma_{x'J'F'; xJF} \\ = \sigma_{x'J'; xJ}^{I=0} \frac{(2F'+1)}{(2J'+1)(2I+1)}. \end{aligned} \quad (12b)$$

As shown in the introduction, these relations guarantee that, as long as all hyperfine states have equal excitation temperature, only the corresponding pure rotational cross sections are relevant for determining the excitation conditions.

3. CS and IOS-approximation

Computing the transition matrix elements between rotational states in their general form as used in (8), usually is not possible with a reasonable amount of computing time and memory

capacity; it has been done in some special cases for comparison with simpler approximations (Green, 1976, 1980). Commonly used approximations are, first, the CS (coupled states) approximation, introduced by McGuire and Kouri (1974) and then the IOS (infinite order sudden) approximation (Pack, 1974; Goldflam et al., 1977, and references therein). These approximations are usually formulated for the system of coupled differential equations for the wave function resulting from the Schrödinger equation. The formulation in the framework of formal scattering theory presented here gives precise insight into the underlying assumptions for these approximations. The calculations which follow are done for the case of a symmetric top molecule. This includes the case of a linear rotor when $K=0$. The extensions necessary for treating the inversion motion of NH_3 are easy to handle (Appendix A). Generalizing the treatment for the case of an asymmetric top molecule is straightforward.

It is most convenient to formulate the scattering equations in the helicity representation, originally introduced by Jacob and Wick (1959), for the description of scattering involving spin-polarized particles. In this representation, instead of the magnitude of the orbital angular momentum of the relative motion l , the scattering states are described by the projection λ_J of the total angular momentum of the scattering molecule J on the direction of the relative motion k . As the projection of the orbital angular momentum l on the direction of the relative motion is 0, λ_J is also the projection of the total angular momentum $L=J+l$.

Formulating the scattering equations in the helicity representation is equivalent to formulating the corresponding Schrödinger equations in a molecule-fixed coordinate system, as has been done by several authors (Walker and Light, 1975; McGuire and Kouri, 1974; Pack, 1974). The transformation coefficients between the total angular momentum representation $|k, LM_L, J\rangle$ and the helicity representation $|k, LM_L, J\lambda_J\rangle$ are (Jacob and Wick, 1959, Appendix B)

$$\begin{aligned} \langle kLM_L J\lambda_J | kLM_L J\lambda_J \rangle \\ = (2l+1)^{1/2} (-1)^{l-J+\lambda_J} \begin{pmatrix} l & J & L \\ 0 & \lambda_J & -\lambda_J \end{pmatrix}. \end{aligned} \quad (13a)$$

Applying the transformation to the potential matrix elements (7) gives the potential matrix in the helicity representation

$$\begin{aligned} \langle k'J'K'\lambda_J' | V^L | kJK\lambda_J \rangle \\ = \sum_{\lambda_\mu} \sum_{l'l'} [ll']^{1/2} (-1)^{l-J+\lambda_J+l'-J'+\lambda_J'} \\ \cdot \begin{pmatrix} l & J & L \\ 0 & \lambda_J & -\lambda_J \end{pmatrix} \begin{pmatrix} l' & J' & L \\ 0 & \lambda_J' & -\lambda_J' \end{pmatrix} (-1)^{K+L+J+J'} \left(\frac{[J'J'l\lambda_J]}{4\pi} \right)^{1/2} \\ \cdot \begin{pmatrix} l' & \lambda & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} J' & \lambda & J \\ -K' & -\mu & K \end{pmatrix} \left\{ \begin{matrix} J' & \lambda & J' \\ l & L & l' \end{matrix} \right\} v_{\lambda_\mu}^{l'l}(k', k). \end{aligned} \quad (13b)$$

The aim of the CS-approximation now is to diagonalize the potential matrix in λ_J ; the transition matrix given by the Lippmann-Schwinger equation (4) is then also diagonal in λ_J as the collision energy and thereby the Green operator do not depend on λ_J . The diagonalization is achieved by approximating the magnitude of the orbital angular momentum l in (13b) by the magnitude of the total angular momentum $L=J+l=J'+l'$ in the radial matrix element of the potential:

$$v_{\lambda_\mu}^{l'l}(k', k) \sim v_{\lambda_\mu}^{LL}(k', k). \quad (14)$$

[In the original formulation by McGuire and Kouri (1974) l^2 is replaced by L^2 in the Schrödinger equation for the radial

wavefunction.] The physical reasons for this approximation are discussed in detail by McGuire (1976). Insertion of (14) in (13b) allows the calculation of the l, l' -sum and finally one gets

$$\begin{aligned} \langle k'J'K'\lambda_J' | V^L | kJK\lambda_J \rangle = \delta_{\lambda_J\lambda_J'} \sum_{\lambda_\mu} (-1)^{K+\lambda_J} \left(\frac{[J'J\lambda_J]}{4\pi} \right)^{1/2} \\ \cdot \begin{pmatrix} J' & \lambda & J \\ -K' & -\mu & K \end{pmatrix} \begin{pmatrix} J' & \lambda & J \\ \lambda_J & 0 & -\lambda_J \end{pmatrix} v_{\lambda_\mu}^{LL}(k', k). \end{aligned} \quad (15)$$

Let $T_{\dots;\dots}^{L\lambda_J}$ be the diagonal transition matrix elements

$$T_{\dots\lambda_J;\dots\lambda_J}^L = T_{\dots;\dots}^{L\lambda_J} \delta_{\lambda_J\lambda_J'}$$

and correspondingly $V_{\dots;\dots}^{L\lambda_J}$ are the diagonal potential matrix elements (15). The Lippmann-Schwinger equation (4) then reads

$$\begin{aligned} T_{k'J'K';kJK}^{L\lambda_J} = V_{k'J'K';kJK}^{L\lambda_J} + \frac{1}{(2\pi)^3} \int k''^2 dk'' \sum_{J''K''} \\ \cdot V_{k'J'K';k''J''K''}^{L\lambda_J} \frac{1}{E - E_{k''J''K''} + i\eta} T_{k''J''K'';kJK}^{L\lambda_J}, \end{aligned} \quad (16)$$

which is one set of integral equations for each value of λ_J (the corresponding set of differential equations for the wave function decouples into one set of equations for each specific value of λ_J). Based on this approximation, Green (1981) has done extensive calculations for the collision systems He/NH_3 and He/HCN and showed the CS-approximation to be accurate to about 10% in general and to about 50% in the extreme cases.

A much more crude approximation is the IOS-approximation, where the energy-splitting of the molecular energy levels is neglected in comparison to the collision energy by setting

$$E_{k''J''K''} \sim E_{k''}. \quad (17)$$

The Lippmann-Schwinger equation (16) then is solved by the Ansatz [corresponding to (15)]

$$\begin{aligned} T_{k'J'K';kJK}^{L\lambda_J} = \sum_{AM} T_{AM}^L(k', k) (-1)^{K+\lambda_J} \left(\frac{[J'J\lambda_J]}{4\pi} \right)^{1/2} \\ \cdot \begin{pmatrix} J' & A & J \\ -K' & -M & K \end{pmatrix} \begin{pmatrix} J' & A & J \\ \lambda_J & 0 & -\lambda_J \end{pmatrix}, \end{aligned} \quad (18)$$

which can be easily verified by a straightforward calculation, resulting in the Lippmann-Schwinger equation for the T_{AM}^L

$$\begin{aligned} T_{AM}^L(k', k) = v_{AM}^{LL}(k', k) \\ + \frac{1}{(2\pi)^3} \cdot \int k''^2 dk'' (-1)^M \sum_{\tilde{A}\tilde{M}} \left(\frac{[A\tilde{A}\lambda_J]}{4\pi} \right)^{1/2} \\ \sum_{\lambda_\mu} v_{\lambda_\mu}^{LL}(k', k'') T_{\tilde{A}\tilde{M}}^L(k'', k) \frac{1}{E - E_{k''} + i\eta} \\ \cdot \begin{pmatrix} \lambda & \tilde{A} & A \\ \mu & M & -M \end{pmatrix} \begin{pmatrix} \lambda & \tilde{A} & A \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned} \quad (19)$$

The physical meaning of the T_{AM}^L becomes clear by multiplying both sides of (19) with $Y_{AM}(\Omega)$ and summing over A, M . Making use of the addition theorem for the spherical harmonics the result is

$$\begin{aligned} T_{k'k}^L(\Omega) = v_{k'k}^{LL}(\Omega) \\ + \frac{1}{(2\pi)^3} \cdot \int k''^2 dk'' v_{k'k''}^{LL}(\Omega) \frac{1}{E - E_{k''} + i\eta} T_{k''k}^L(\Omega), \end{aligned}$$

i.e., an 1-dimensional scattering equation for each direction $\Omega = (\theta, \phi)$ of the relative motion with the effective potential

$$V_{\Omega}(R) = v(R, \Omega) + \frac{\hbar^2 L(L+1)}{2\mu R^2}.$$

The second term is the centrifugal potential for a scattering state with total angular momentum L . Within the IOS-approximation, calculation of the transition matrix is reduced to the integration of a 1-dimensional Schrödinger equation for each direction of the relative motion. The “on the energy shell” transition matrix elements necessary for calculating the cross sections are obtained by setting $k=k'$, as the rotational energies of the scattering molecules are neglected.

The T_{AM}^L are calculated in a way similar to the one described by Green (1979). Making use of the orthogonality and completeness relations for the transformation coefficients (13a) and Eq. (17) the transition matrix in the CS- and IOS-approximation is given by

$$T_{K'J'K'v';kJKI}^L = \sum_{AM} (-1)^{K+A+L} \left(\frac{[l'JJ'A]}{4\pi} \right)^{1/2} T_{AM}^L(k', k) \cdot \begin{pmatrix} J' & A & J \\ K' & M & -K \end{pmatrix} \begin{pmatrix} l & A & l' \\ 0 & 0 & 0 \end{pmatrix} \left\{ \begin{matrix} l & A & l' \\ J' & L & J \end{matrix} \right\}. \quad (20)$$

As shown in Appendix A the inversion states of NH_3 may be included by replacing the matrix elements according to (A.6). Inserting (20) into (12) and including the inversion states we finally get the hyperfine selective collision cross sections for a symmetric top molecule

$$\begin{aligned} \sigma_{J'K'F'v';JKFv}(k) &= [J'JF]^2 \frac{\pi}{k^2} \sum_{\substack{\tilde{L}\tilde{L}' \\ AA'}} (-1)^{\tilde{L}+\tilde{L}'} [\tilde{L}\tilde{L}'] ([AA'])^{1/2} \\ &\cdot \frac{1}{4\pi} f(AA', \tilde{L}\tilde{L}', JJ', FF', vv', KK'; I) \\ &\cdot n_K^2 n_{K'}^2 \left\{ \begin{pmatrix} J' & A' & J \\ K' & M_- & -K \end{pmatrix} \bar{T}_{A'M_-}^{\tilde{L}'}(k) - v(-1)^J \right. \\ &\cdot \left. \begin{pmatrix} J' & A' & J \\ K' & -M_+ & K \end{pmatrix} \bar{T}_{A'-M_+}^{\tilde{L}'}(k) \right\}^* \\ &\cdot \left\{ \begin{pmatrix} J' & A & J \\ K' & M_- & -K \end{pmatrix} \bar{T}_{AM_-}^{\tilde{L}}(k) - v(-1)^J \right. \\ &\cdot \left. \begin{pmatrix} J' & A & J \\ K' & -M_+ & K \end{pmatrix} \bar{T}_{A-M_+}^{\tilde{L}}(k) \right\}, \end{aligned} \quad (21)$$

where $\bar{T}_{AM}^{\tilde{L}}$ is an abbreviation for $2\pi i(k\mu/\hbar^2) T_{AM}^L$, $M_- = K - K'$, $M_+ = K + K'$ and

$$\begin{aligned} f(AA', \tilde{L}\tilde{L}', JJ', FF', vv', KK'; I) &= \sum_{l'l'} [l'lL] \frac{1}{2} (1 + vv'(-1)^{l+l'+K+K'}) \\ &\cdot \begin{pmatrix} l & l' & A \\ 0 & 0 & 0 \end{pmatrix} \left\{ \begin{matrix} l & A & l' \\ J' & \tilde{L} & J \end{matrix} \right\} \left\{ \begin{matrix} \tilde{L} & I & L \\ F & l & J \end{matrix} \right\} \left\{ \begin{matrix} \tilde{L} & I & L \\ F' & l' & J' \end{matrix} \right\} \\ &\cdot \begin{pmatrix} l & l' & A' \\ 0 & 0 & 0 \end{pmatrix} \left\{ \begin{matrix} l & A' & l' \\ J' & \tilde{L}' & J \end{matrix} \right\} \left\{ \begin{matrix} \tilde{L}' & I & L \\ F & l & J \end{matrix} \right\} \left\{ \begin{matrix} \tilde{L}' & I & L \\ F' & l' & J' \end{matrix} \right\}. \end{aligned} \quad (22)$$

For the case of a linear rotor (e.g., HCN) set $K=K'=0$, $v(-1)^J = -1$; then (21) reduces to

$$\begin{aligned} \sigma_{J'F';JF}(k) &= [J'JF]^2 \frac{\pi}{k^2} \sum_{\substack{\tilde{L}\tilde{L}' \\ AA'}} (-1)^{\tilde{L}+\tilde{L}'} [\tilde{L}\tilde{L}'] ([AA'])^{1/2} \\ &\cdot \frac{1}{4\pi} f(AA', \tilde{L}\tilde{L}', JJ', FF', -(-1)^J - (-1)^{J'}, 00; I) \\ &\cdot \begin{pmatrix} J' & A & J \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} J' & A' & J \\ 0 & 0 & 0 \end{pmatrix} \bar{T}_{A'0}^{\tilde{L}'}(k) \bar{T}_{A0}^{\tilde{L}}(k). \end{aligned} \quad (23)$$

4. Numerical results

Calculation of the scattering phase shifts necessary for determining the IOS transition matrix elements is done in WKB approximation. The WKB phase shifts are calculated very efficiently by a Gauss-Mehler quadrature (Pack, 1974; Cohen, 1978). In this way the $T^L(\Omega)$ are computed for a set of angular points θ, ϕ ; their spherical harmonic expansion coefficients are derived by numerical integration.

The hyperfine selective cross sections then are computed by combining the T_{AM}^L with the corresponding Wigner 3j- and 6j-symbols according to (21), (22). Whereas the IOS-approximation shows typical errors of 20% (Green, 1979) at collision energies of 100–200 K, it is much more inaccurate at lower energies. The relative factors

$$g_{F'F}(x'J', xJ; k) = \frac{\sigma_{x'J'F', xJF}(k)}{\sigma_{x'J', xF}(k)} \quad (24)$$

calculated in IOS-approximation, however, show only small variations with the collision energy in contrast to the absolute value of the IOS cross sections (see Tables 1 and 3). It is therefore a reasonable good approximation to combine the relative factors (24) calculated in IOS-approximation with the absolute values of the rotational cross sections calculated in more accurate approximations (e.g. CS). Only for the transitions within the hyperfine multiplet of one rotation-inversion state, the (elastic) cross sections are not known from CS-calculations. For these transitions the IOS-cross sections have to be taken directly for the hyperfine selective collisional excitation. For these quasielastic processes, however, the IOS-approximation should be rather accurate, as the energy transfer involved is extremely small compared to the collision energy.

In Table 1 the relative factors for the collision system para NH_3/He are given at four different collision energies. The potential used is the same as the one Green (1980) used for the CS-calculations. The quasi-elastic hyperfine selective cross sections calculated in IOS-approximation are given in Table 2. The corresponding data for the scattering system HCN/He are given in Tables 3 and 4. The HCN/He potential is taken from Green and Thaddeus, 1974. Only the lower $\rightarrow$ upper rates are given in the tables, the corresponding down rates may be calculated with the details balance relations. For NH_3 only the upper inversion levels are taken into account as final states; as may be seen from Eq. (21) the rates with the lower inversion levels as final states are then given by

$$\begin{aligned} \sigma_{J', K', F', v' = -1; J, K, F, v} &= \sigma_{J', K', F', v' = +1; J, K, F, -v} \quad (\text{para } \text{NH}_3). \end{aligned}$$

Table 1. The relative factors [Eq. (24)] of the hyperfine selective collision rates for para NH_3 in IOS approximation; the four values (from top to bottom) are for a kinetic temperature of 15, 25, 50, and 100 K

JKF	J' K' F' U	1			2			2		
		1	2	1	1	3	2	1	3	1
110L			0.27	0.73				0.26	0.15	0.59
			0.30	0.70				0.28	0.15	0.57
			0.31	0.69				0.30	0.15	0.55
			0.35	0.65				0.34	0.15	0.51
2	5.40E-02	0.64	0.30					0.24	0.68	8.39E-02
	5.98E-02	0.63	0.31					0.24	0.67	8.56E-02
	6.22E-02	0.63	0.31			*)		0.25	0.67	8.61E-02
	6.94E-02	0.61	0.32					0.25	0.66	8.86E-02
1	0.24	0.51	0.25					0.51	0.22	0.26
	0.23	0.52	0.25					0.50	0.23	0.27
	0.23	0.52	0.25					0.49	0.24	0.27
	0.22	0.53	0.25					0.46	0.26	0.28
110U	5.31E-02	0.14		6.27E-02	0.84	0.10	0.68	0.25	7.71E-02	
	3.26E-02	0.11		6.48E-02	0.83	0.11	0.65	0.27	7.84E-02	
	2.48E-02	8.61E-02		6.72E-02	0.82	0.11	0.71	0.22	7.38E-02	
	1.37E-02	4.41E-02		3.81E-02	0.86	9.91E-02	0.56	0.35	9.23E-02	
2		5.86E-02	0.31	0.31	0.38	0.35	0.50	0.15		
		4.30E-02	0.31	0.31	0.38	0.35	0.50	0.16		
		3.27E-02	0.31	0.32	0.38	0.35	0.51	0.14		
		1.72E-02	0.32	0.29	0.39	0.35	0.47	0.18		
1			6.09E-02	0.60	0.33	0.20	0.48	0.32		
			6.40E-02	0.60	0.33	0.20	0.48	0.31		
			6.69E-02	0.60	0.33	0.19	0.48	0.34		
			5.41E-02	0.62	0.32	0.23	0.49	0.27		
221L			0.46	0.15	0.39	0.15	0.78	6.60E-02		
			0.43	0.17	0.40	0.17	0.75	7.51E-02		
			0.41	0.18	0.41	0.18	0.73	8.32E-02		
			0.36	0.21	0.43	0.18	0.73	8.38E-02		
3			6.54E-02	0.70	0.23	0.38	0.29	0.33		
			7.39E-02	0.68	0.25	0.38	0.30	0.32		
			7.57E-02	0.67	0.25	0.37	0.31	0.31		
			8.94E-02	0.64	0.28	0.37	0.31	0.31		
2			0.23	0.33	0.44	0.38	0.53	9.14E-02		
			0.24	0.35	0.41	0.37	0.53	0.10		
			0.25	0.36	0.39	0.37	0.52	0.11		
			0.26	0.39	0.36	0.37	0.52	0.11		
221U			4.21E-02	0.11	0.49	0.42	9.27E-02			
			2.51E-02	9.20E-02	0.47	0.43	9.83E-02			
			1.91E-02	7.13E-02	0.44	0.45	0.10			
			1.20E-02	4.14E-02	0.45	0.47	8.89E-02			
3			6.77E-02	0.38	0.44	0.18				
			5.08E-02	0.38	0.43	0.19				
			3.92E-02	0.38	0.43	0.19				
			2.31E-02	0.38	0.42	0.20				
2			0.17	0.54	0.29					
			0.19	0.53	0.28					
			0.20	0.53	0.27					
			0.20	0.54	0.27					
212L			0.40	0.36	0.24					
			0.38	0.37	0.24					
			0.37	0.39	0.25					
			0.34	0.42	0.24					
3			0.26	0.66	8.43E-02					
			0.27	0.64	9.47E-02					
			0.28	0.62	0.10					
			0.30	0.56	0.14					
1			0.40	0.20	0.40					
			0.40	0.22	0.37					
			0.41	0.24	0.35					
			0.39	0.32	0.29					
212U			9.23E-02	6.76E-02						
			6.92E-02	5.46E-02						
			5.29E-02	4.22E-02						
			2.73E-02	2.25E-02						
3				1.71E-02						
				9.75E-03						
				7.03E-03						
				3.05E-03						
1										

Table 1 (continued)

JKF	J' K' F'	3 2 3 U			3 1 3 U			4 4 3 U		
		4	2		4	2		5	4	
110L		0.13	0.78	8.84E-02	0.67	0.22	0.11	0.15	0.24	0.61
		0.15	0.75	0.10	0.60	0.27	0.13	0.17	0.29	0.54
		0.16	0.73	0.11	0.59	0.28	0.13	0.18	0.32	0.50
		0.18	0.72	0.10	0.64	0.24	0.12	0.18	0.36	0.45
2		0.36	0.28	0.36	0.36	0.45	0.19	0.23	0.42	0.36
		0.36	0.30	0.35	0.35	0.44	0.21	0.24	0.41	0.35
		0.36	0.31	0.34	0.35	0.44	0.21	0.25	0.41	0.35
		0.36	0.31	0.34	0.35	0.45	0.20	0.26	0.40	0.35
1		0.36	0.55	9.21E-02	0.18	0.46	0.35	0.35	0.44	0.20
		0.35	0.54	0.10	0.21	0.46	0.33	0.33	0.44	0.23
		0.35	0.53	0.11	0.22	0.46	0.32	0.31	0.44	0.25
		0.35	0.53	0.12	0.20	0.46	0.34	0.29	0.44	0.27
110U		0.66	0.23	0.11	0.22	0.11	0.66	0.68	0.11	0.21
		0.63	0.25	0.13	0.24	0.12	0.64	0.65	0.12	0.23
		0.59	0.27	0.13	0.20	9.57E-02	0.70	0.62	0.13	0.25
		0.64	0.24	0.12	0.25	0.22	0.53	0.68	7.57E-02	0.25
2		0.35	0.45	0.19	0.26	0.64	9.93E-02	0.12	0.61	0.27
		0.35	0.45	0.20	0.26	0.63	0.10	0.12	0.60	0.28
		0.35	0.44	0.21	0.25	0.66	9.01E-02	0.13	0.59	0.28
		0.35	0.45	0.20	0.28	0.57	0.15	0.11	0.62	0.27
1		0.19	0.46	0.35	0.49	0.18	0.33	0.36	0.17	0.47
		0.20	0.46	0.34	0.48	0.19	0.33	0.36	0.18	0.46
		0.22	0.46	0.33	0.51	0.16	0.33	0.36	0.20	0.45
		0.20	0.46	0.34	0.45	0.26	0.29	0.37	0.17	0.46
221L		0.27	0.13	0.59	0.45	0.42	0.13			
		0.28	0.14	0.58	0.42	0.44	0.14			
		0.29	0.14	0.57	0.39	0.46	0.15			
		0.31	0.13	0.55	0.33	0.52	0.15			
3		0.23	0.69	8.53E-02	0.38	0.38	0.24			
		0.23	0.68	8.95E-02	0.37	0.38	0.25			
		0.23	0.68	8.92E-02	0.36	0.38	0.26			
		0.24	0.67	8.67E-02	0.36	0.36	0.28			
2		0.52	0.24	0.24	0.20	0.50	0.30			
		0.51	0.25	0.24	0.23	0.49	0.28			
		0.50	0.25	0.25	0.26	0.48	0.26			
		0.47	0.27	0.26	0.30	0.47	0.23			
221U		0.40	0.15	0.45	0.49	0.24	0.27	8.06E-02	0.75	0.17
		0.40	0.17	0.43	0.48	0.25	0.27	9.01E-02	0.72	0.19
		0.40	0.13	0.47	0.47	0.27	0.26	0.10	0.69	0.21
		0.39	0.27	0.35	0.49	0.27	0.24	0.10	0.68	0.21
3		0.28	0.62	0.10	0.34	0.51	0.15	0.43	0.22	0.34
		0.29	0.60	0.11	0.34	0.50	0.16	0.42	0.23	0.34
		0.28	0.64	8.85E-02	0.35	0.49	0.17	0.40	0.25	0.35
		0.31	0.53	0.16	0.35	0.48	0.17	0.40	0.25	0.35
2		0.37	0.33	0.30	0.22	0.43	0.34	0.12	0.46	0.42
		0.35	0.34	0.30	0.23	0.44	0.34	0.13	0.46	0.41
		0.37	0.32	0.31	0.23	0.44	0.33	0.15	0.46	0.39
		0.33	0.38	0.28	0.21	0.45	0.34	0.16	0.46	0.39
212L		0.20	0.50	0.30	0.58	0.20	0.22	0.35	0.42	0.23
		0.23	0.49	0.28	0.56	0.22	0.23	0.33	0.42	0.26
		0.26	0.48	0.26	0.55	0.22	0.23	0.31	0.42	0.28
		0.30	0.47	0.23	0.51	0.25	0.24	0.29	0.43	0.29
3		0.38	0.38	0.24	0.20	0.73	7.02E-02	0.19	0.46	0.35
		0.37	0.38	0.25	0.21	0.71	7.60E-02	0.20	0.45	0.35
		0.36	0.38	0.26	0.22	0.71	7.87E-02	0.22	0.44	0.34
		0.36	0.36	0.28	0.23	0.68	8.75E-02	0.25	0.41	0.35
1		0.45	0.42	0.13	0.23	0.11	0.66	0.28	0.26	0.46
		0.42	0.44	0.14	0.24	0.12	0.64	0.28	0.29	0.43
		0.39	0.46	0.15	0.25	0.12	0.62	0.27	0.32	0.41
		0.33	0.52	0.15	0.27	0.14	0.59	0.24	0.38	0.38
212U		0.25	0.42	0.33	0.37	0.34	0.29	0.34	0.33	0.33
		0.25	0.42	0.32	0.37	0.35	0.28	0.34	0.34	0.33
		0.26	0.43	0.31	0.38	0.33	0.29	0.33	0.35	0.32
		0.26	0.44	0.31	0.30	0.40	0.30	0.35	0.34	0.31
3		0.34	0.51	0.15	0.29	0.59	0.12	0.12	0.58	0.30
		0.34	0.50	0.16	0.29	0.57	0.14	0.13	0.56	0.31
		0.34	0.49	0.17	0.28	0.61	0.11	0.14	0.55	0.31
		0.34	0.47	0.18	0.32	0.52	0.16	0.12	0.57	0.31
1		0.47	0.24	0.29	0.38	0.19	0.43	0.45	0.14	0.40
		0.46	0.26	0.28	0.37	0.23	0.40	0.44	0.16	0.41
		0.44	0.29	0.27	0.38	0.17	0.44	0.42	0.18	0.41
		0.44	0.31	0.25	0.41	0.27	0.32	0.44	0.14	0.42

*)

Table 1 (continued)

JKF	J'	K'	F'									
	3	2	3	4	2	3	4	2	3	4	5	4
323L	0.56	0.24	0.20			0.28	0.45	0.27	0.14	0.43	0.43	
	0.50	0.28	0.22			0.29	0.45	0.26	0.16	0.43	0.41	
	0.46	0.30	0.24			0.30	0.44	0.25	0.18	0.43	0.39	
	0.39	0.35	0.26			0.34	0.44	0.22	0.19	0.43	0.39	
4	0.19	0.76	5.06E-02			0.35	0.41	0.23	0.45	0.21	0.34	
	0.21	0.72	6.13E-02			0.35	0.42	0.23	0.43	0.23	0.34	
	0.23	0.70	6.39E-02			0.35	0.41	0.24	0.40	0.26	0.34	
	0.27	0.62	0.11			0.34	0.38	0.28	0.38	0.27	0.34	
2	0.28	9.11E-02	0.63			0.37	0.42	0.21	8.56E-02	0.73	0.19	
	0.31	0.11	0.58			0.36	0.42	0.22	0.10	0.69	0.21	
	0.34	0.11	0.54			0.36	0.43	0.21	0.12	0.65	0.23	
	0.37	0.19	0.44			0.31	0.50	0.19	0.14	0.62	0.25	
323U		8.76E-02	7.15E-02			0.28	0.40	0.31	0.30	0.48	0.22	
		6.66E-02	5.74E-02			0.28	0.41	0.31	0.29	0.47	0.24	
		5.12E-02	4.43E-02			0.27	0.42	0.31	0.27	0.46	0.27	
		2.73E-02	2.41E-02			0.27	0.42	0.31	0.24	0.44	0.32	
4			1.93E-02			0.31	0.58	0.11	0.31	0.31	0.39	
			1.08E-02			0.32	0.57	0.12	0.30	0.32	0.38	
			8.02E-03			0.32	0.54	0.13	0.31	0.32	0.37	
			3.71E-03			0.33	0.53	0.14	0.34	0.31	0.36	
2						0.44	0.20	0.36	0.12	0.48	0.40	
						0.44	0.21	0.35	0.14	0.48	0.38	
						0.43	0.24	0.33	0.15	0.49	0.36	
						0.43	0.26	0.32	0.15	0.54	0.31	
313L						0.52	0.27	0.22	0.34	0.42	0.24	
						0.46	0.30	0.24	0.33	0.42	0.25	
						0.41	0.32	0.27	0.32	0.42	0.26	
						0.33	0.38	0.29	0.30	0.43	0.27	
4						0.21	0.74	5.73E-02	0.16	0.49	0.35	
						0.23	0.70	6.94E-02	0.18	0.48	0.35	
						0.25	0.68	6.98E-02	0.19	0.46	0.35	
						0.30	0.58	0.12	0.23	0.42	0.35	
2						0.31	0.10	0.59	0.32	0.24	0.44	
						0.34	0.12	0.54	0.31	0.27	0.43	
						0.37	0.13	0.50	0.30	0.29	0.41	
						0.40	0.22	0.38	0.26	0.35	0.39	
313U						8.82E-02	7.20E-02	0.31	0.31	0.33	0.36	
						6.79E-02	5.85E-02	0.31	0.31	0.34	0.34	
						5.19E-02	4.51E-02	0.32	0.32	0.36	0.32	
						2.75E-02	2.44E-02	0.33	0.33	0.37	0.30	
4							1.94E-02	9.70E-02	0.62	0.29	0.29	
							1.11E-02	0.10	0.60	0.29	0.30	
							7.83E-03	0.12	0.58	0.30	0.31	
							3.59E-03	0.12	0.57	0.31	0.31	
2									0.48	0.14	0.38	
									0.47	0.14	0.39	
									0.44	0.17	0.40	
									0.42	0.16	0.42	
443L									0.66	9.12E-02	0.25	
									0.62	0.11	0.27	
									0.60	0.11	0.29	
									0.53	0.14	0.33	
5									5.80E-02	0.76	0.18	
									6.84E-02	0.73	0.21	
									6.97E-02	0.71	0.22	
									9.13E-02	0.66	0.25	
4									0.19	0.22	0.58	
									0.21	0.25	0.54	
									0.22	0.26	0.51	
									0.26	0.31	0.43	
443U									3.21E-02	9.58E-02		
									1.84E-02	7.65E-02		
									1.31E-02	5.94E-02		
									9.70E-03	3.94E-02		
5										7.08E-02		
										5.44E-02		
										4.19E-02		
										2.81E-02		
4												

*) The IOS-rates are identically 0 for these transitions.

Table 2. The IOS hyperfine selective collision rates within one hyperfine multiplet for para NH_3/He . Units are cm^3s^{-1} . [The rates are the same for both inversion levels]

J K	F → F'	T			
		15 K	25 K	50 K	340 K
1 1	0 2	3.57E-11	2.49E-11	2.04E-11	1.05E-11
	0 1	9.32E-11	8.66E-11	7.07E-11	3.37E-11
	2 1	3.94E-11	3.28E-11	2.69E-11	1.31E-11
2 2	1 3	2.79E-11	1.92E-11	1.56E-11	9.18E-12
	1 2	7.59E-11	7.02E-11	5.84E-11	3.18E-10
	3 2	4.48E-11	3.88E-11	3.21E-11	1.77E-11
2 1	2 3	6.09E-11	5.26E-11	4.32E-11	2.08E-11
	2 1	4.46E-11	4.14E-11	3.45E-11	1.72E-11
	3 1	1.13E-11	7.41E-12	5.74E-12	2.32E-12
3 2	3 4	5.70E-11	4.96E-11	4.14E-11	2.07E-11
	3 2	4.66E-11	4.28E-11	3.58E-11	1.83E-11
	4 2	1.26E-11	8.06E-12	6.49E-12	2.82E-12
3 1	3 4	5.76E-11	5.09E-11	4.21E-11	2.09E-11
	3 2	4.71E-11	4.38E-11	3.66E-11	1.86E-11
	4 2	1.27E-11	8.30E-12	6.35E-12	2.74E-12
4 4	3 5	2.07E-11	1.37E-11	1.06E-11	7.45E-12
	3 4	6.17E-11	5.70E-11	4.82E-11	3.03E-11
	5 4	4.56E-11	4.05E-11	3.40E-11	2.16E-11

(*) the rates are the same for both inversion levels

Table 4. Same as Table 2, but for HCN/He

J	F → F'	T			
		15 K	25 K	50 K	100 K
1	1 2	8.50E-11	8.10E-11	6.79E-11	5.66E-11
	1 0	3.21E-11	3.29E-11	2.33E-11	1.45E-11
	2 0	1.20E-11	1.09E-11	1.04E-11	1.03E-11
2	2 3	7.25E-11	7.27E-11	5.78E-11	4.71E-11
	2 1	5.15E-11	5.20E-11	4.08E-11	3.07E-11
	3 1	1.49E-11	1.45E-11	1.19E-11	1.04E-11
3	3 4	6.60E-11	6.69E-11	5.27E-11	4.21E-11
	3 2	5.35E-11	5.50E-11	4.36E-11	3.35E-11
	4 2	1.50E-11	1.42E-11	1.09E-11	1.04E-11
4	4 5	6.01E-11	6.20E-11	4.88E-11	3.87E-11
	4 3	5.20E-11	5.41E-11	4.31E-11	3.36E-11
	5 3	1.38E-11	1.33E-11	9.57E-12	8.74E-12

Table 3. Same as Table 1, but for HCN/He ; the temperatures are from top to bottom 15, 25, 50, and 100 K

J' F'	2			3			4		
	2	3	1	3	4	2	4	5	3
J F									
	1 1	0.34	0.43	0.23	0.42	0.28	0.29	0.36	0.33
		0.37	0.39	0.23	0.41	0.31	0.28	0.38	0.31
		0.36	0.40	0.24	0.42	0.29	0.29	0.37	0.33
		0.36	0.40	0.24	0.43	0.30	0.28	0.36	0.35
2	2	0.32	0.51	0.17	0.29	0.55	0.16	0.32	0.48
		0.31	0.54	0.16	0.30	0.53	0.17	0.31	0.50
		0.31	0.53	0.16	0.29	0.54	0.16	0.32	0.47
		0.31	0.53	0.16	0.29	0.54	0.17	0.32	0.46
0	0	0.38	0.37	0.25	0.28	0.24	0.48	0.33	0.29
		0.35	0.35	0.31	0.28	0.28	0.44	0.32	0.26
		0.38	0.33	0.29	0.27	0.26	0.47	0.32	0.30
		0.37	0.34	0.29	0.25	0.28	0.47	0.33	0.31
2 2	2 2				0.36	0.38	0.27	0.45	0.29
					0.39	0.35	0.26	0.41	0.31
					0.38	0.36	0.26	0.44	0.29
					0.38	0.36	0.26	0.45	0.29
3	3				0.31	0.53	0.17	0.28	0.56
					0.29	0.56	0.15	0.29	0.56
					0.29	0.55	0.16	0.28	0.57
					0.30	0.54	0.16	0.27	0.57
1	1				0.36	0.29	0.35	0.28	0.24
					0.34	0.25	0.41	0.31	0.23
					0.35	0.27	0.39	0.29	0.21
					0.35	0.28	0.38	0.28	0.22
3 3	3 3							0.37	0.36
								0.41	0.33
								0.39	0.34
								0.39	0.34
4	4							0.30	0.53
								0.28	0.56
								0.29	0.56
								0.29	0.55
2	2							0.35	0.26
								0.32	0.24
								0.33	0.23
								0.33	0.24

5. Discussion

In the simplest approximation used in the literature for the hyperfine selective collisional excitation the hyperfine selective cross sections are taken to be proportional to the statistical weights of the final states, an approximation which at least guarantees that the rates agree with the detailed balance relations (see, e.g., Guilloteau and Baudry, 1981). Tables 1–4 clearly show that this approximation is much too simple.

The expression for the transition matrix in IOS-approximation (20) is very similar to the expression (7) obtained for the potential matrix elements by expanding the angular dependence in a multipole series. The only difference is that the orbital angular momentum quantum numbers l, l' occurring in the potential matrix elements $v^{l',l}(k', k)$ are replaced by the total angular momentum L in the expansion coefficients of the transition matrix $T_{AM}^L(k', k)$. The reason is the corresponding replacement $v^{l',l} \sim v^{LL}$ made in deriving the CS-approximation.

As shown by Green (1979) the simple structure of (20) results in the expression

$$\begin{aligned} \sigma_{J'K'v';JKv} &= (n_K n_{K'})^2 [J] \sum_A \frac{1}{2} (1 + vv'(-1)^{A+K+K'}) \\ &\cdot \left\{ \begin{pmatrix} J' & A & J \\ K' & M_- & K \end{pmatrix}^2 Q(A, M_-, M_-) \right. \\ &+ \begin{pmatrix} J' & A & J \\ K' & -M_+ & K \end{pmatrix}^2 Q(A, M_+, M_+) \\ &- v(-1)^J \begin{pmatrix} J' & A & J \\ K' & M_- & K \end{pmatrix} \begin{pmatrix} J' & A & J \\ K' & -M_+ & K \end{pmatrix} \\ &\cdot [Q(A, M_-, -M_+) + Q(A, -M_+, M_-)] \Big\}, \end{aligned} \quad (25)$$

where

$$Q(A, M_1, M_2) = \frac{\pi}{k^2} \sum_L (2L+1) \frac{1}{4\pi} \bar{T}_{AM_2}^{L*} \bar{T}_{AM_1}^L. \quad (26)$$

For the case of a linear rotor ($K = K' = 0$, $v(-1)^J = -1$) (24) reduces to an expression with the same structure as the one derived by Varshalovich and Khersonsky (1977), which they obtained starting from a multipole expansion of the scattering matrix. Such an expansion in general is not possible, as the nonlinearity of the Lippmann-Schwinger equation always mixes the contributions from different orders of multipoles. A multipole expansion is possible, however, when the scattering problem is handled in first order perturbation theory (i.e., $T \sim V$); instead of the expression (20) derived in IOS-approximation, the expansion of the T -matrix then has exactly the form of the expression (7) for the expansion of the potential matrix elements.

The difference between (20) and (7) has no influence on the resulting expression for the cross sections (24); only the meaning of the Q -factors is different, when obtained by the IOS- or the multipole expansion approximation. The difference is important however, when one calculates the hyperfine selective cross sections. As the hyperfine selective transition matrix elements for a specific total angular momentum L contain contributions from rotational transition matrix elements for the total angular momentum $\tilde{L}$ where $|L-I| < \tilde{L} < L+I$, the hyperfine selective cross sections contain mixing terms of the form $T^{\tilde{L}*} T^L$, which cannot be combined into a formula corresponding to (25), (26). This is only possible, when one assumes that in addition to the IOS-

approximation $T^{\tilde{L}'} \sim T^{\tilde{L}} \sim T^L$. Equation (21) then reduces to

$$\begin{aligned} \sigma_{J'K'F'v';JKFv} &= (n_K n_{K'})^2 [J'JF] \sum_A \frac{1}{2} (1 + vv'(-1)^{A+K+K'}) \\ &\cdot \left\{ \begin{pmatrix} F' & A & F \\ J & I & J' \end{pmatrix}^2 \left\{ \begin{pmatrix} J' & A & J \\ K' & M_- & K \end{pmatrix}^2 Q(A, M_-, M_-) \right. \right. \\ &+ \begin{pmatrix} J' & A & J \\ K' & -M_+ & K \end{pmatrix}^2 Q(A, M_+, M_+) \\ &- v(-1)^J \begin{pmatrix} J' & A & J \\ K' & M_- & K \end{pmatrix} \begin{pmatrix} J' & A & J \\ K' & M_+ & K \end{pmatrix} \\ &\cdot [Q(A, M_-, -M_+) + Q(A, -M_+, M_-)] \Big\}. \end{aligned} \quad (27)$$

For the case of a linear rotor this again reduces to the expression derived by Varshalovich and Khersonsky (1977) for hyperfine selective collision cross sections. Without the additional approximation $T^{\tilde{L}'} \sim T^{\tilde{L}} \sim T^L$ however, the more complicated formula (23) is the appropriate one.

As an aside let us note that from (27) we immediately get the formula for the radiative rates between hyperfine states: for a specific multipole transition order A we retain only one multipole term in the transition matrix, i.e., from (26) $Q(A', \dots) = \delta_{A,A'} Q(A, \dots)$. Comparing (25) and (27) we see, that the hyperfine radiative rates (Einstein A -coefficients) are related to the rotational rates by the well known formula

$$A_{J'F' \dots, JF \dots} = (2J+1)(2F+1) \left\{ \begin{pmatrix} F' & A & F \\ J & I & J' \end{pmatrix}^2 A_{J' \dots, J \dots} \right.$$

Tables 5 and 6, where the hyperfine selective relative factors $g_{FF'}$ calculated according to (26) and (27) are compared for some selected transitions for the case of a symmetric top and a linear rotor molecule, show that the formula given by Varshalovich and Khersonsky is indeed too simple an approximation for the calculation of hyperfine selective collisional rates. In some cases, transition rates, that vanish in the multipole-expansion approximation, have an intensity even comparable to the other hyperfine transitions in the IOS-approximation.

The importance of the correct description of the hyperfine selective collisional excitation is presented in Fig. 1. It shows the relative intensity of the outer hyperfine satellites of the $\text{NH}_3(1, 1)$ inversion transition as a result of radiative transfer calculations. Two different models for the hyperfine selective collisional excitation were used: one taking the hyperfine rates being proportional to the statistical weights of the final levels, the other taking the correct hyperfine selective rates as calculated in this paper. The results for both models substantially differ. This demonstrates that a correct interpretation of the observational data requires the precise knowledge of the hyperfine selective collisional excitation.

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Table 5. Relative factors of the hyperfine selective collision rates in IOS- (upper) and multipole-expansion- (lower) approximation; para NH_3/He ; Kin. temp. 340 K

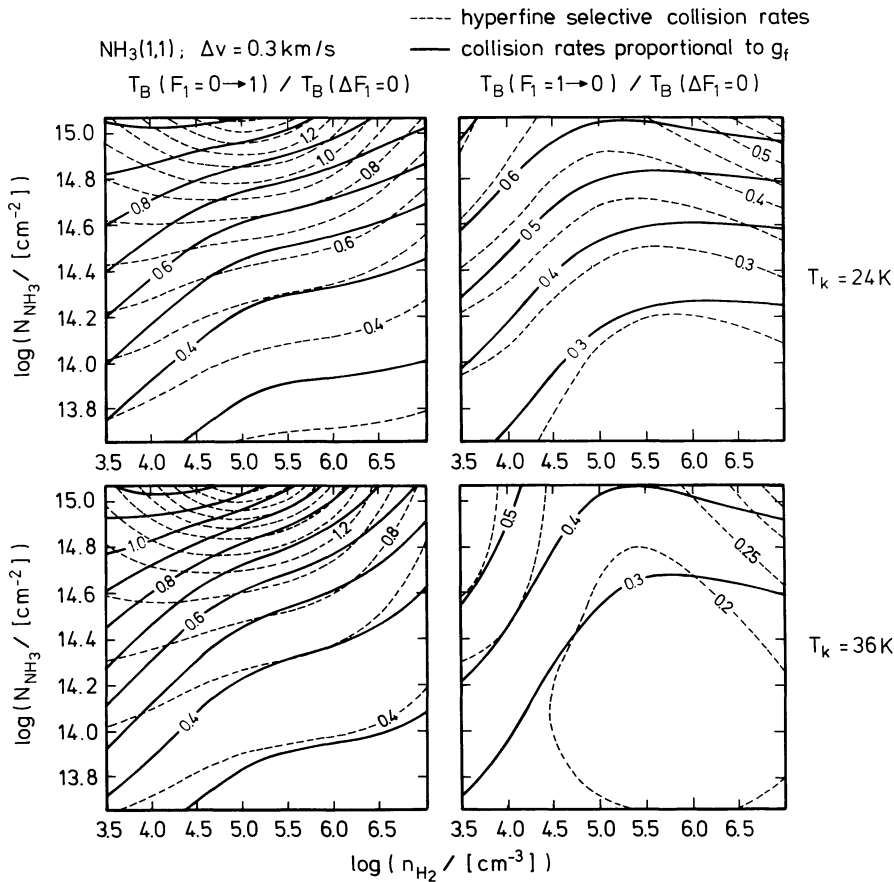
$J' \begin{smallmatrix} K' \\ F' \end{smallmatrix}$	1 1 0 U	2 2 1	3 3 1
JKF			
110L	0.35 0.00	0.65 1.00	0.34 0.00 0.15 0.08 0.51 0.92
2	0.07 0.00	0.61 0.75	0.32 0.25 0.25 0.17 0.66 0.80 0.09 0.03
1	0.22 0.33	0.53 0.42	0.25 0.25 0.46 0.72 0.26 0.03 0.28 0.25
110U			0.56 1.00 0.35 0.00 0.09 0.00
2			0.35 0.35 0.47 0.56 0.18 0.09
1			0.23 0.08 0.49 0.47 0.27 0.45
212L			0.34 0.47 0.42 0.30 0.24 0.23
3			0.30 0.22 0.56 0.74 0.14 0.05
1			0.39 0.39 0.32 0.11 0.29 0.50

Table 6. Relative factors of the hyperfine selective collision rates in IOS- (upper) and multipole-expansion- (lower) approximation; HCN/He ; Kin. temp. 100 K

$J' \begin{smallmatrix} K' \\ F' \end{smallmatrix}$	2 2	3 3	1 1	3 3	4 4	2 2
JF						
11	0.36 0.60	0.40 0.24	0.24 0.16	0.43 0.61	0.30 0.12	0.28 0.27
2	0.31 0.24	0.53 0.62	0.16 0.14	0.29 0.23	0.54 0.66	0.17 0.11
0	0.37 0.00	0.34 0.36	0.29 0.64	0.25 0.00	0.28 0.19	0.47 0.81
22				0.38 0.65	0.36 0.18	0.26 0.17
3				0.30 0.19	0.54 0.72	0.16 0.09
1				0.35 0.15	0.28 0.17	0.38 0.68

Appendix A: the inversion motion of NH_3

There has been some confusion in the literature about the orientation of the molecule fixed coordinates for the description of NH_3 resulting in inconsistent phase factors for the wave functions. Green (1980) (Appendix), discusses these errors and introduces a consistent notation, which will be used here.

**Fig. 1.** The intensity ratio of the outer hyperfine satellites ($F=0 \rightarrow 1$, $F=1 \rightarrow 0$) of the NH_3 (J, K) = (1, 1) inversion transition relative to the main line ($\Delta F=0$) obtained from radiative transfer calculations. The relative intensities are plotted as a contour map in the $\log N_{\text{NH}_3} / \log n_{\text{H}_2}$ plane (N_{NH_3} is the column density of NH_3 , n_{H_2} is the volume density of H_2) for two different kinetic temperatures T_k of the model cloud. The radiative transfer model used is described in detail in Stutzki (1984) and Stutzki and Winnewisser (1984). The results are plotted for two different models of the hyperfine selective collisional excitation: 1) the dashed lines give the results obtained with the collision rates calculated in this paper, 2) the solid lines presents the results calculated with the simple assumption that the collision rates are proportional to the statistical weights of the final hyperfine states

The rotational eigenfunctions of NH_3 with the right symmetry properties in regard to the hydrogen nuclei are given by

$$|JKM\varepsilon\rangle = n_K \sqrt{\frac{1}{6}} \{ |JKM\rangle (\beta\alpha\alpha + \alpha\beta\alpha e^{2\pi i K/3} + \alpha\alpha\beta e^{4\pi i K/3}) \\ + \varepsilon |J-KM\rangle (\beta\alpha\alpha + \alpha\beta\alpha e^{-2\pi i K/3} + \alpha\alpha\beta e^{-4\pi i K/3}) \}$$

with $n_K = (1 + \delta_{K,0})^{1/2}$, $K \geq 0$, $\varepsilon = \pm 1$ (for $K=0$ only $+1$). The normalization factor n_K , which is important only for $K=0$, is not included by Green (1980).

Due to the threefold symmetry of NH_3 the matrix elements between rotational states of some operator A which is diagonal in the hydrogen spin quantum numbers can be evaluated to give

$$\langle J'K'M'\varepsilon | A | JK M \varepsilon \rangle \\ = \frac{1}{2} n_K n_{K'} \{ (\langle J'K'M' | + \varepsilon \langle J'-K'M' |) A \\ \cdot (|JKM\rangle + \varepsilon |J-KM\rangle) \}. \quad (\text{A.1})$$

The wave functions $|v\rangle$ describing the inversion motion can be approximated by the sum and difference of a wave function centered at the equilibrium position of the ^{14}N -nucleus h_e

$$|v\rangle = \sqrt{\frac{1}{2}} \{ f(h-h_e) + v f(h+h_e) \}; \quad v = \pm 1,$$

where $v = +1$ is the lower and $v = -1$ is the upper inversion state. The symmetry of the total rotational-inversion wave function

$$|JKMv\rangle = |JKM\varepsilon\rangle |v\rangle \quad (\text{A.2})$$

with respect to interchanging two H atoms then requires

$$v\varepsilon(-1)^J = -1 \quad (\text{A.3})$$

(Green, 1980). The matrix elements between inversion states can be calculated by approximating the inversion wave function $f(h)$ by a δ -function (Davis and Boggs, 1978) resulting in

$$\langle v' | A(h) | v \rangle = \frac{1}{2} \{ A(h_e) + v' A(-h_e) \}. \quad (\text{A.4})$$

The rotational matrix elements of $A(-h_e)$ by symmetry considerations can be shown to give

$$\langle \dots K' \dots l' | A(-h_e) | \dots K \dots l \rangle \\ = (-1)^{l+l'+K+K'} \langle \dots | A(h_e) | \dots \rangle. \quad (\text{A.5})$$

Combining (A.1), (A.4), and (A.5) we thus finally get for the matrix elements between rotation inversion states

$$\langle k'J'K'\nu'l' | A | kJK\nu l \rangle \\ = \frac{1}{2} (1 + \nu\nu'(-1)^{l+l'+K+K'}) n_K n_{K'} \\ \cdot \{ \langle k'J'K'l' | A | kJKl \rangle - \nu(-1)^J \langle k'J'K'l' | A | kJ-Kl \rangle \}, \quad (\text{A.6})$$

where we dropped the argument h_e from the operator A .

Note added in proof: After submission of this article a new paper on HCN hyperfine selective collisional excitation by T. Monteiro (1984, *Monthly Notices Roy. Astron. Soc.* **211**, 257) was published.

Monteiro numerically solved the set of differential equations for the wave function obtained by applying a CS-approximation to the total angular momentum states, i.e. by replacing $l^2 = (L-F)^2 = L^2 + F^2 - 2L \cdot F$ with $\tilde{L}^2$. In contrast, we apply the CS approximation to the states where the nuclear spin is already decoupled. The remaining angular momentum is $\tilde{L} = L - I$ and the CS approximation results in replacing $l^2 = (\tilde{L} - J)^2 = \tilde{L}^2 + J^2 - 2\tilde{L} \cdot J$ with $\tilde{L}^2$. By comparing $\tilde{L}^2 = (L-I)^2 = L^2 + I^2 - 2L \cdot I$ with $\tilde{L}^2$ one sees that Monteiro completely neglects the (kinematical) $L \cdot I$ coupling responsible for the occurrence of $T^{\tilde{L}} T^{\tilde{L}}$ interference terms in our

exact Eq. (11). This kinematical coupling, however, is the essential part in the hyperfine selective collision process, as the potential itself is diagonal in all nuclear spin variables.

As a result of this too simplistic approximation Monteiro obtains “selection rules” not valid in general, but similar to the Varshalovich and Khersonsky approach (Tables 5 and 6). In his calculations the $J=1$, $F=0 \rightarrow J'$, $F'=J'$ rates for example vanish identically. The fact, that Monteiro finds the $J=0$, $F=1 \rightarrow J'$, F' rates to follow the final state degeneracies only approximately, is due to the poor numerical accuracy ($\sim 25\%$) of his calculations; his “approximate” result may be shown to be exact by evaluating the $J=0$, $F=1 \rightarrow J'F'$ cross sections directly from our exact Eq. (11).

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